

LAB 4 Setup Description and Guide

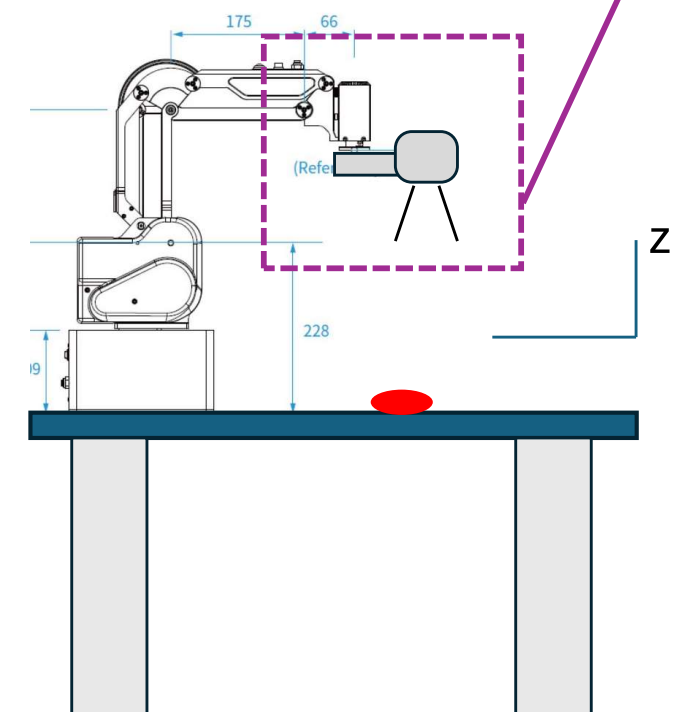
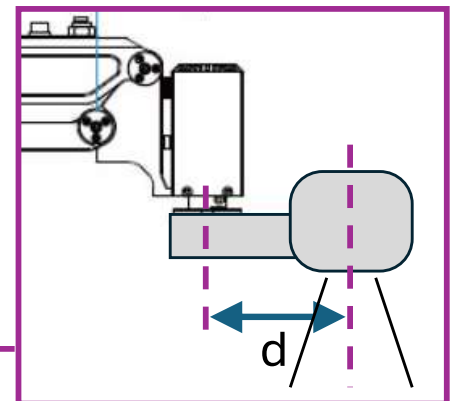
In this lab, you will implement and test a visual servoing system. A webcam has been attached to the end effector of each robot arm, facing downward. During operation, you will use feedback from the camera to control the motion of the robot arm. Therefore, please connect your laptop to both the robot and the webcam.

The robot can translate the camera along all three axes of the inertial frame and rotate it about the z-axis shown in the figure. Note that the camera lens may not be aligned with the axis of rotation of the robot's last joint. This offset is denoted by d in the figure below. You need to measure the parallel distance between the camera's optical axis and the axis of rotation of J4.

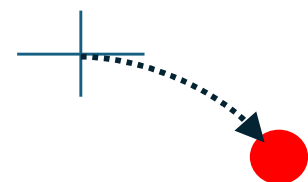
The camera may not be properly calibrated. Therefore, please calibrate the camera each time you begin working with a setup. The calibration process requires a printed checkerboard pattern with ArUco markers. You can generate this board using the board generator code used in the lectures, *get_aruco_board.py*, which has been uploaded to SUCourse. Print the generated board and place it in front of the camera. A sample calibration code was presented in class, and you may use it for this step. First, capture 20 images of the checkerboard at various positions and orientations, and store them in a designated folder. This can be done using *capture_charuco_images.py*. Then, run *calibrate_charuco.py* to obtain the camera calibration matrix.

Make sure that the captured images represent the expected operating conditions of the lab. The best way to achieve this is to move the camera to different positions and orientations using the robot arm, by commanding the robot to various joint configurations.

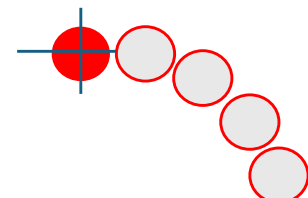
Note: Inside '*get_aruco_board.py*' some board settings (number of squares, size of the squares and markers) are configured by you. Those settings should be entered to '*calibrate_charuco.py*' too. The marker and square dimensions may alter when you print the generated image. Make sure the printed dimensions are entered to the calibration code.



Camera view



Camera view



Tasks:

Once the calibration is complete, you can implement the visual servoing task. In this lab, there is a single task to complete. An object detection function provided by us, `find_red_object` inside `myCode.py`, will return the position of a red object in the pixel frame. Please use a flat marker approximately the size of a bottle cap as the red object.

Your task is to use this function to obtain marker position feedback and move the robot accordingly. The goal is to bring the marker to the center of the camera view by moving the camera with the robot, as illustrated in the figure below. Your code should work for various initial robot arm configurations and for different marker locations, provided that the marker is initially visible within the camera view.

Note that the camera does not provide direct depth measurements. However, the depth can be inferred from the position of the end-effector and its vertical distance from the tabletop.

Deliverables:

Please submit your code, sample plots showing the trajectory of the marker in the pixel frame, and a sample video recorded by the webcam. The video should clearly show the marker being moved to the center of the camera view.

In your report, use the plots to analyze the performance of your system for different marker locations. In particular, compare the system behavior when the marker is placed far from the robot's origin with cases where the marker is placed near the robot's home position or other representative locations.

**Don't forget to check out the camera calibration
and object detection codes provided on SUCourse!**